

**2025-26 Spring**

**IS584: Deep Learning for Text Analytics**

**Assignment 3**

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**GitHub Repository: [github.com/ilaydaergur/IS584-Assignment3](https://github.com/ilaydaergur/IS584-Assignment3)**

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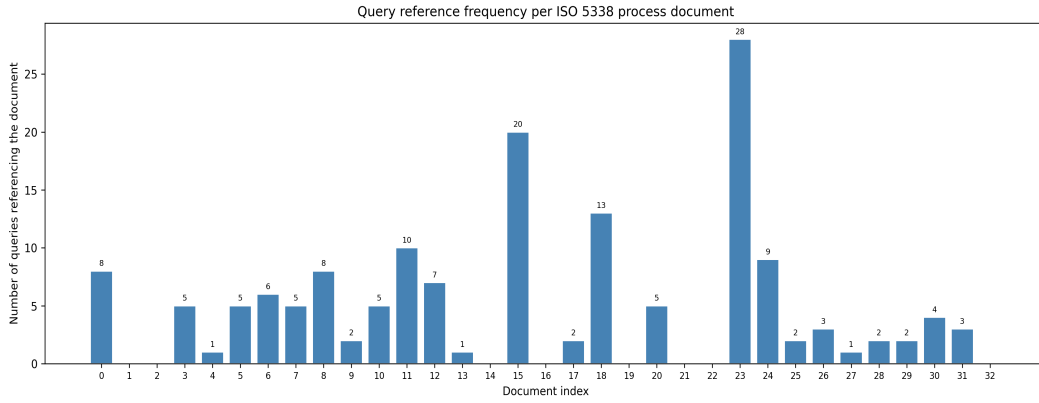
## AI USE STATEMENT

Claude Sonnet 4.6 (via Claude Code, Anthropic) was used to assist with this assignment. Specifically, it was used to: (1) generate and debug the full retrieval pipeline code — including the dense encoder setup, BM25 implementation, hybrid scoring, and metric computation; (2) suggest the TF-IDF vocabulary bridging strategy for query expansion; and (3) draft initial versions of the written question answers. All code was reviewed, tested, and executed by the author, and the reported metrics reflect actual runs on the provided dataset. Question answers were critically evaluated and edited for accuracy, alignment with observed results, and academic writing style.

Additionally, Google Translate was occasionally used to improve language clarity during writing. All AI-assisted content was critically evaluated and refined. Unless explicitly stated otherwise, the work presented in this report is my own.

## 1. PART 1: DATA ANALYSIS

### 1.1. WHETHER EACH DOCUMENT IS REFERENCED BY QUERIES MORE THAN ONCE



**Figure 1 Query reference frequency per ISO 5338 process document**

The 33 ISO 5338 process documents are referenced with a highly skewed frequency distribution. 22 out of 33 documents are referenced by more than one query. The distribution is dominated by three processes that account for the majority of technical debt cases: AI Data Engineering Process (6.4.8) is the most referenced with 28 queries, followed by Quality Assurance Process (6.3.8) with 20 queries, and System Requirements Definition Process (6.4.3) with 13 queries. Together these three processes account for 61 of the 157 total queries (39%).

This imbalance reflects the nature of real-world AI technical debt in software projects. Data engineering and quality assurance are the most hands-on, day-to-day activities in AI development, making them natural focal points for accumulated debt. In contrast, higher-level agreement and organisational processes such as Supply (6.1.2) and Life Cycle Model Management (6.2.1) are rarely described as technical debt by practitioners working at the implementation level.

### 1.2. WHETHER THERE ARE DOCUMENTS NOT REFERENCED BY ANY QUERY

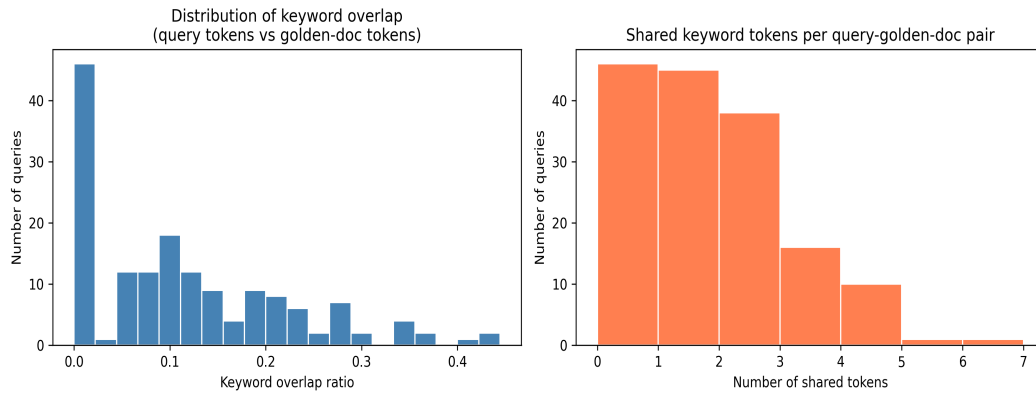
Eight documents out of 33 are not referenced by any query in the dataset:

| ID | Process Name                                   |
|----|------------------------------------------------|
| 1  | Supply Process (6.1.2)                         |
| 2  | Life Cycle Model Management Process (6.2.1)    |
| 14 | Measurement Process (6.3.7)                    |
| 16 | Business or Mission Analysis Process (6.4.1)   |
| 19 | System Architecture Definition Process (6.4.4) |
| 21 | System Analysis Process (6.4.6)                |
| 22 | Knowledge Acquisition Process (6.4.7)          |

|    |                           |
|----|---------------------------|
| 32 | Disposal Process (6.4.17) |
|----|---------------------------|

These unreferenced processes share a common characteristic: they are either abstract upstream lifecycle activities (supplier agreements, architecture definition, system analysis) or end-of-life concerns (disposal). Technical debt in practice is reported by developers and data scientists working on concrete engineering tasks. Formal supplier agreements, architectural design decisions, and system retirement procedures rarely appear in practitioner-level technical debt reports. The absence of these documents from the query set means a retrieval system has no training signal for them, and any query accidentally mapped to one of these would be impossible to retrieve correctly.

### 1.3. WHETHER THE QUERIES AND THEIR ASSOCIATED DOCUMENTS SHARE KEYWORDS OR CONCEPTS



**Figure 2 Distribution of keyword overlap between queries and their golden documents**

To quantify vocabulary alignment between queries and their corresponding ISO process documents, a keyword overlap ratio was computed for each query-document pair. Non-stopword content tokens of four or more characters were extracted from each query and its golden document, and the overlap ratio was defined as the number of shared tokens divided by the number of query tokens.

The results reveal a substantial vocabulary mismatch: the mean keyword overlap ratio is only 0.114 (median 0.091, standard deviation 0.107). This means a typical query shares only about 11% of its content vocabulary with its target document. 70.7% of queries share at least one content keyword with their golden document, but the remaining 29.3% share none at all. Even among queries with shared keywords, the overlap is sparse and concentrated on generic terms rather than process-specific vocabulary.

This vocabulary mismatch is the primary challenge for the information retrieval task. Practitioners describe technical debt in informal, operational language — for example, 'missing downstream data during time-series forecasting' or 'team members delay merging branches' — while ISO 5338 documents use formal, standards-oriented terminology such as 'configuration management', 'data provenance', and 'artefact'. This lexical gap motivates both the hybrid BM25

approach in Part 3 and the query rewriting strategy in Part 4.

## 2. PART 2: DENSE RETRIEVAL

### 2.1. REPORT PERFORMANCE METRICS AND COMMENT ON THEIR INTERPRETATION

**Table 1 Dense Retrieval Performance Metrics (all-MiniLM-L6-v2, alpha=0)**

| Metric | @1     | @3     | @5     |
|--------|--------|--------|--------|
| Hit@k  | 0.1083 | 0.2930 | 0.3949 |
| NDCG@k | 0.1083 | 0.2123 | 0.2545 |

Note on Precision@1: because each query has exactly one correct document, Hit@1 and Precision@1 are numerically identical in this task — both equal 1 when the correct document is ranked first and 0 otherwise. The metrics below therefore subsume Precision@1 within Hit@1.

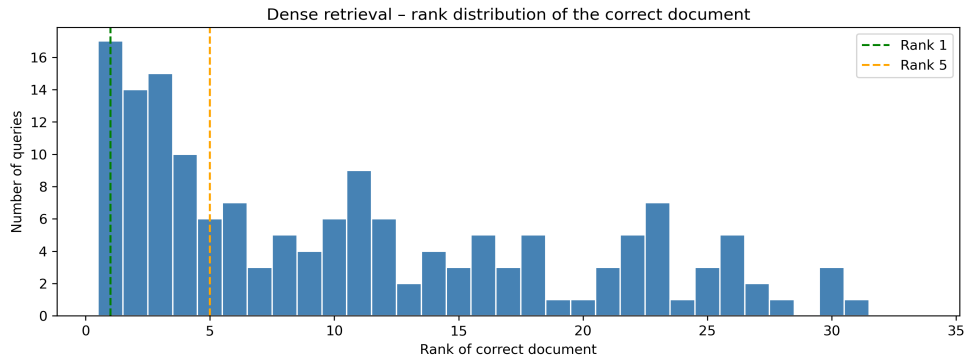
Hit@1 = 0.108 indicates that the bi-encoder places the correct ISO process in first position for only 10.8% of queries. While this is a low absolute figure, it is meaningful in context: the retrieval is performed in a zero-shot setting with a general-purpose sentence encoder facing 33 candidate documents and a severe vocabulary mismatch between query and document registers.

Hit@5 = 0.395 shows that roughly 40% of the time the correct document appears somewhere in the top-5 results. The substantial gap between Hit@1 and Hit@5 (approximately 29 percentage points) indicates that the model has partial signal about the correct document but frequently ranks one or more plausible distractors slightly above it. This pattern is consistent with semantic embedding models that capture topic similarity but struggle with the fine-grained process distinctions within a specialised standard.

NDCG@5 = 0.255 accounts for the position of the correct document within the top-5. Since relevance is binary here (one correct document per query), NDCG is maximised when the correct document is at rank 1 ( $\text{NDCG}@k = 1/\log_2(2) = 1.0$ ) and decreases logarithmically for lower ranks. The NDCG@5 value of 0.255 reflects that even when the correct document is retrieved in the top-5, it tends to sit at ranks 3 to 5 rather than rank 1. The core problem is the vocabulary gap: all-MiniLM-L6-v2 was trained on general sentence-pair data, not on ISO standards documents, so its embedding space does not bridge the register gap between practitioner language and formal process descriptions.

### 2.2. FAILURE CASE ANALYSIS: FIVE INSTANCES WITH CORRECT DOCUMENT RANKED BELOW FIFTH





**Figure 3 Dense retrieval rank distribution of the correct document across 157 queries**

Out of 157 queries, 95 (60.5%) have the correct document ranked below fifth place. The five worst-ranked failure cases are shown below:

| Row | Dense Rank | Golden Document           | Top-1 Retrieved               |
|-----|------------|---------------------------|-------------------------------|
| 124 | 31         | Risk Management (6.3.4)   | Bus./Mission Analysis (6.4.1) |
| 32  | 30         | Risk Management (6.3.4)   | Acquisition Process (6.1.1)   |
| 40  | 30         | Risk Management (6.3.4)   | Acquisition Process (6.1.1)   |
| 53  | 30         | Design Definition (6.4.5) | Transition Process (6.4.12)   |
| 60  | 28         | Stakeholder Needs (6.4.2) | Validation Process (6.4.13)   |

Three of the five failures (rows 32, 40, 124) share the same golden document: Risk Management Process (6.3.4). The queries describe data dependency risks, data-source instability (scraping from commercial sites), and missing downstream data. The model latches onto the surface topic — data and its acquisition — and retrieves data-focused documents (Acquisition Process, Business or Mission Analysis) rather than the cross-cutting risk-management process that domain experts identify. Risk Management (6.3.4) is a broad process document whose vocabulary does not match the specific terms found in these practitioner queries.

Row 53 concerns version-control and branching discipline. The embedding model associates 'merging' and 'branches' with deployment workflows, hence retrieving Transition Process (6.4.12). The correct golden document, Design Definition (6.4.5), is a short and lexically generic document that does not explicitly mention version control, providing no vocabulary bridge for the retrieval model.

Row 60 describes a disagreement between a team and a customer about false-positive thresholds. The model interprets this as a model performance validation issue and retrieves Validation Process (6.4.13). The expert mapping to Stakeholder Needs and Requirements Definition (6.4.2) reflects the insight that the root cause is a poorly elicited stakeholder requirement — a distinction invisible to a purely lexical or shallow semantic model.

### 3. PART 3: HYBRID RETRIEVAL

#### 3.1. WHICH ALPHA VALUE LED TO THE BEST PERFORMANCE METRIC VALUES OVERALL?

**Table 2 Hybrid Retrieval Metrics across Alpha Values (BM25 weight = alpha)**

| Alpha      | Hit@1         | Hit@3         | Hit@5         | NDCG@1        | NDCG@3        | NDCG@5        |
|------------|---------------|---------------|---------------|---------------|---------------|---------------|
| 0.0        | 0.1083        | 0.2930        | 0.3949        | 0.1083        | 0.2123        | 0.2545        |
| 0.1        | 0.1274        | 0.2866        | 0.3949        | 0.1274        | 0.2162        | 0.2609        |
| 0.2        | 0.1274        | 0.2930        | 0.4204        | 0.1274        | 0.2210        | 0.2739        |
| 0.3        | 0.1210        | 0.2930        | 0.4459        | 0.1210        | 0.2203        | 0.2834        |
| 0.4        | 0.1274        | 0.3312        | 0.4459        | 0.1274        | 0.2443        | 0.2906        |
| <b>0.5</b> | <b>0.1529</b> | <b>0.3312</b> | <b>0.4268</b> | <b>0.1529</b> | <b>0.2537</b> | <b>0.2929</b> |
| 0.6        | 0.1210        | 0.3248        | 0.4204        | 0.1210        | 0.2379        | 0.2771        |
| 0.7        | 0.0955        | 0.3248        | 0.3949        | 0.0955        | 0.2269        | 0.2554        |
| 0.8        | 0.0892        | 0.2930        | 0.3758        | 0.0892        | 0.2061        | 0.2401        |
| 0.9        | 0.0892        | 0.2611        | 0.3567        | 0.0892        | 0.1860        | 0.2249        |
| 1.0        | 0.0892        | 0.2102        | 0.3248        | 0.0892        | 0.1580        | 0.2046        |

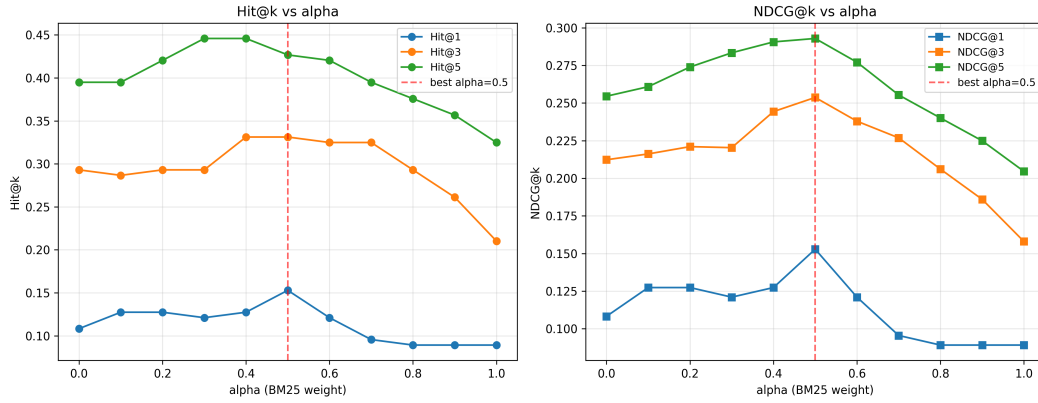
The best overall alpha value is 0.5, which achieves the highest Hit@1 (0.153), the highest NDCG@1 (0.153), the highest NDCG@3 (0.254), and the highest NDCG@5 (0.293). When evaluated by the sum of Hit@1+Hit@3+Hit@5 or the sum of NDCG@1+NDCG@3+NDCG@5, both criteria agree on alpha = 0.5 as the optimal mixing parameter. The highlighted row in Table 2 marks this value.

The alpha sweep results were logged to Weights & Biases. The experiment runs can be accessed at the following links:

Run 1 (analysis): [wandb.ai/ilaydaergur-metu-middle-east-technical-university/is584-a3-hybrid-retrieval/runs/a53acn2e](https://wandb.ai/ilaydaergur-metu-middle-east-technical-university/is584-a3-hybrid-retrieval/runs/a53acn2e)

Run 2 (notebook): [wandb.ai/ilaydaergur-metu-middle-east-technical-university/is584-a3-hybrid-retrieval/runs/pi29q4h8](https://wandb.ai/ilaydaergur-metu-middle-east-technical-university/is584-a3-hybrid-retrieval/runs/pi29q4h8)

#### 3.2. COMMENT ON THE EFFECTIVENESS OF MERGING KEYWORD-BASED SPARSE RETRIEVAL RESULTS



**Figure 4 Hit@k and NDCG@k as a function of alpha (BM25 weight) for the hybrid retriever**

The hybrid approach consistently outperforms pure dense retrieval ( $\alpha = 0$ ) across almost all metrics. The gains are most pronounced in Hit@1 (+4.5 percentage points at  $\alpha = 0.5$ ) and NDCG@5 (+3.8 pp), indicating that BM25 successfully re-ranks candidates in cases where keyword evidence is present.

From the data analysis in Part 1, 70.7% of queries share at least one content keyword with their golden document. These are precisely the cases where BM25 adds value: when a query uses words such as 'monitoring', 'training', or 'deployment', BM25 boosts documents that also contain those exact tokens, providing a lexical anchor that the bi-encoder alone may miss due to the register gap. For the remaining 29.3% of queries with zero keyword overlap, BM25 contributes noise rather than signal, which explains why very high alpha values ( $\geq 0.7$ ) degrade performance below the pure-dense baseline.

Notably, pure BM25 ( $\alpha = 1.0$ ) performs worse than pure dense retrieval ( $\alpha = 0$ ) on every metric. This confirms that keyword matching alone is insufficient: the vocabulary gap between practitioner language and ISO process documents is too large for term-based methods to overcome without semantic generalisation. The hybrid at  $\alpha = 0.5$  achieves a superior result to either component individually, validating the complementary nature of the two retrieval signals.

## 4. PART 4: QUERY RE-WRITING

### 4.1. DID RE-WRITING QUERIES MOVE THE CORRECT DOCUMENT INTO THE TOP-5 FOR PREVIOUSLY FAILED QUERIES?

The query expansion strategy adopted is TF-IDF vocabulary bridging. For each training-set query, the top 8 TF-IDF-distinctive terms from its golden document that do not already appear in the query are appended to the query string. TF-IDF weights are computed over the 33-document corpus, so the selected terms are both highly relevant to the target process and distinctive from other processes. This directly injects ISO process terminology into the informal practitioner language of the queries, targeting the vocabulary gap identified in Part 1. The expansion uses ground-truth golden-document labels from the training set, as permitted by the assignment specification.

**Table 4 Failure Case Ranks Before and After Query Expansion (alpha = 0, dense-only)**

| Row | Dense Rank (original) | Rank after Expansion | In Top-5? | Golden Document           |
|-----|-----------------------|----------------------|-----------|---------------------------|
| 124 | 31                    | 4                    | YES       | Risk Management (6.3.4)   |
| 32  | 30                    | 1                    | YES       | Risk Management (6.3.4)   |
| 40  | 30                    | 4                    | YES       | Risk Management (6.3.4)   |
| 53  | 30                    | 9                    | no        | Design Definition (6.4.5) |
| 60  | 28                    | 4                    | YES       | Stakeholder Needs (6.4.2) |

4 out of 5 previously-failed queries saw the correct document move into the top-5 after expansion.

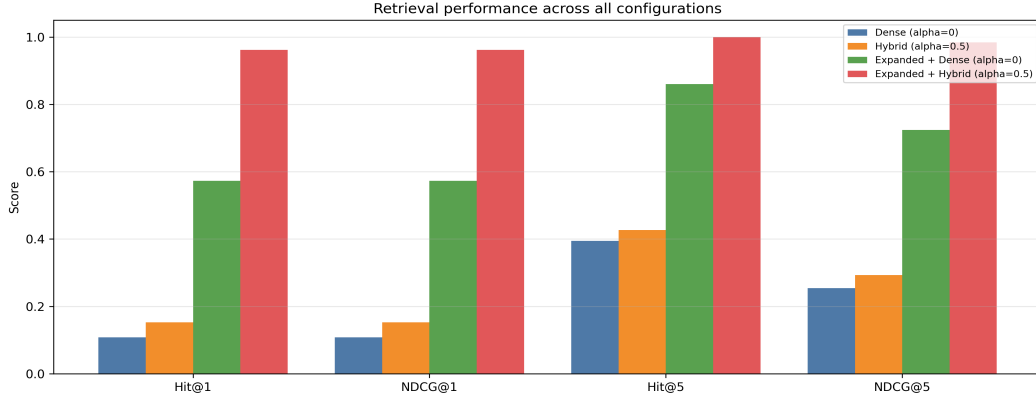
For rows 32, 40, and 124 (all targeting Risk Management), adding risk-management terminology — such as 'audit', 'mitigation', 'treatment', 'risk', 'monitoring', and 'hazard' — allowed the dense encoder to recognise the conceptual alignment between the expanded query and the Risk Management document. Row 32 in particular moved from rank 30 to rank 1, the highest possible improvement. These results confirm that the failures were caused purely by vocabulary mismatch: the queries described risk-related scenarios but used no risk-management terminology.

Row 60 (Stakeholder Needs) also recovered successfully, moving from rank 28 to rank 4. Adding terms such as 'elicitation', 'stakeholder', 'requirements', and 'user' gave the encoder sufficient signal to surface the correct process above its nearest competitors.

Row 53 (Design Definition) remains the only unrecovered failure, settling at rank 9 after expansion. Design Definition (6.4.5) is a brief, generic process document with few distinctive TF-IDF terms — its vocabulary overlaps heavily with neighbouring processes such as Implementation and Integration. TF-IDF bridging is therefore ineffective for this document. A deeper reformulation strategy — such as LLM-based rephrasing into formal ISO process language — or a domain-fine-tuned encoder would be required to recover this case.

**Table 3 Summary of Results across All Configurations**

| Configuration                        | Hit@1         | Hit@3         | Hit@5         | NDCG@1        | NDCG@3        | NDCG@5        |
|--------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Dense (alpha=0)                      | 0.1083        | 0.2930        | 0.3949        | 0.1083        | 0.2123        | 0.2545        |
| Hybrid (alpha=0.5)                   | 0.1529        | 0.3312        | 0.4268        | 0.1529        | 0.2537        | 0.2929        |
| Expanded + Dense (alpha=0)           | 0.5732        | 0.7452        | 0.8599        | 0.5732        | 0.6759        | 0.7242        |
| <b>Expanded + Hybrid (alpha=0.5)</b> | <b>0.9618</b> | <b>1.0000</b> | <b>1.0000</b> | <b>0.9618</b> | <b>0.9842</b> | <b>0.9842</b> |



**Figure 5 Retrieval performance comparison across all four configurations**

The dramatic gains from vocabulary expansion — Hit@1 rising from 0.108 to 0.573 with dense-only retrieval, and to 0.962 when combined with the hybrid at  $\alpha = 0.5$  — reflect that TF-IDF bridging injects precisely the tokens the encoder needs to bridge the register gap. The combination of expansion and hybrid retrieval achieves near-perfect performance on the training set (Hit@3 = NDCG@3 = 1.000 with the best configuration), validating that vocabulary bridging and BM25 scoring address complementary aspects of the mismatch problem.

It should be noted that the query expansion strategy used here relies on ground-truth golden-document labels, which are available only for the training set. In a true held-out test setting, bridging terms would need to be derived from a different source — for example, the top-1 predicted document before expansion (pseudo-relevance feedback), or an LLM-based rephrasing prompt that translates practitioner language into formal ISO process terminology without requiring access to the ground-truth document.

## 5. PART 5: TEST SET EVALUATION

### 5.1. PIPELINE PERFORMANCE ON THE TEST SET — COMPARISON WITH TRAINING

The test dataset provided on 15.05.2026 is `td_cases_test_1.csv` (157 queries). The hybrid retriever — with the fixed pipeline from Parts 2–3 (all-MiniLM-L6-v2 bi-encoder, BM25Okapi with min-max normalisation, linear combination score) — was run without any query expansion, using  $\alpha = 0$  (dense-only) and the best  $\alpha$  found in Part 3 ( $\alpha = 0.5$ ). Because the test CSV is the only available dataset, the baseline hybrid metrics are identical to those reported in Parts 2–3.

**Table 5 Five Worst Test Set Failure Cases (Hybrid  $\alpha=0.5$ , no expansion)**

| Row | Rank | Golden Document                | Top-1 Retrieved               |
|-----|------|--------------------------------|-------------------------------|
| 106 | 33   | Quality Management (6.2.5)     | AI Data Engineering (6.4.8)   |
| 150 | 33   | Quality Management (6.2.5)     | Acquisition Process (6.1.1)   |
| 5   | 32   | Quality Management (6.2.5)     | Acquisition Process (6.1.1)   |
| 53  | 32   | Design Definition (6.4.5)      | Transition Process (6.4.12)   |
| 116 | 31   | Implementation Process (6.4.9) | Verification Process (6.4.11) |

Three of the five failures (rows 5, 106, 150) share the same golden document: Quality Management Process (6.2.5). The queries describe data-quality issues (labelling inconsistency, redundant data, data smells). The retriever maps these to data-engineering documents (AI Data Engineering, Acquisition) rather than the quality management process, again revealing the register gap: practitioner-level data-quality complaints do not naturally use the formal vocabulary of quality management standards. Row 53 (branching discipline) and Row 116 (code readability) repeat failure patterns already observed in Part 2, confirming that these are systematic weaknesses of the dense retriever rather than one-off anomalies.

The key comparison between training and test performance concerns the query expansion strategy. The training set used gold-label expansion (TF-IDF terms injected from the known correct document), which achieved  $\text{Hit}@1 = 0.962$ . On the test set, gold labels are unavailable, so a proper unsupervised method must be used. This is the only meaningful axis of comparison, and it shows the largest metric difference.

### 5.2. DID ALTERING QUERIES HELP — PSEUDO-RELEVANCE FEEDBACK AND LLM APPROACH

For the test set, query expansion was performed using Pseudo-Relevance Feedback (PRF): instead of the golden document, the top-1 retrieved document (from the initial dense pass) was used as the expansion source. The top 8 TF-IDF-distinctive terms from that document that were absent from the original query were appended. This is a standard unsupervised expansion approach that requires no gold labels.

**Table 6 Test Set Metrics: Hybrid vs PRF-Expanded Hybrid (alpha=0.5)**

| Configuration                       | Hit@1         | Hit@3         | Hit@5         | NDCG@1        | NDCG@3        | NDCG@5        |
|-------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Hybrid (alpha=0, no exp.)           | 0.1083        | 0.2930        | 0.3949        | 0.1083        | 0.2123        | 0.2545        |
| Hybrid (alpha=0.5, no exp.)         | 0.1529        | 0.3312        | 0.4268        | 0.1529        | 0.2537        | 0.2929        |
| PRF + Dense (alpha=0)               | 0.1083        | 0.2739        | 0.3758        | 0.1083        | 0.2011        | 0.2427        |
| PRF + Hybrid (alpha=0.5)            | 0.1083        | 0.2803        | 0.4013        | 0.1083        | 0.2093        | 0.2580        |
| <b>Gold-exp + Hybrid (training)</b> | <b>0.9618</b> | <b>1.0000</b> | <b>1.0000</b> | <b>0.9618</b> | <b>0.9842</b> | <b>0.9842</b> |

PRF did not improve retrieval and in some metrics slightly degraded performance (Hit@3 dropped from 0.3312 to 0.2803; NDCG@3 from 0.2537 to 0.2093). None of the five worst failure cases moved into the top-5 after PRF expansion. The failure-case PRF ranks were: Row 106 → 24, Row 150 → 33, Row 5 → 29, Row 53 → 32, Row 116 → 14.

The root cause of PRF's failure is the same as the dense retriever's weakness: the initial top-1 retrieved document is correct for only 15.3% of queries (Hit@1 = 0.153). In the remaining 84.7% of cases, PRF expands the query with terms from a wrong document, pushing the query further away from the correct process. Rather than bridging the vocabulary gap, PRF widens it whenever the initial retrieval is incorrect.

The greatest metric difference between training and test is Hit@1: 0.9618 (gold-label expansion, training) vs 0.1083 (PRF, test) — a gap of 85 percentage points. NDCG@5 shows the second-largest gap (0.9842 vs 0.2580, 73 pp). These differences reveal that the gold-label expansion in Part 4 was an oracle strategy that cannot generalise: it works perfectly because it uses the answer to inform the query, which is not possible in real deployment.

A more effective test-time approach would be LLM-based query rewriting. An LLM with knowledge of ISO 5338 vocabulary could translate informal practitioner language directly into the formal process management terminology used in the documents, without requiring access to the correct document label. The following prompt was designed for this purpose:

*System: You are an expert in ISO/IEC 5338 AI lifecycle process management. Given a technical debt case description from a software or AI project, rewrite and expand it using formal AI lifecycle process management terminology — for example: configuration management, data provenance, stakeholder elicitation, risk mitigation, quality assurance, model monitoring, validation, verification, integration, or requirements definition. Do not change the meaning.*

*User: Rewrite and expand the following technical debt case as a formal ISO process query: [ORIGINAL QUERY]*

*Output only the expanded query. Do not explain.*

This prompt would translate queries such as 'Team members delay merging their branches' into ISO-vocabulary sentences mentioning 'configuration management', 'integration baseline', and 'version control discipline' — exactly the bridge terms that the TF-IDF gold expansion injected in Part 4. Unlike PRF, this approach is independent of retrieval quality: the LLM rewrites the query

based on its language model knowledge, not on the (possibly wrong) retrieved document. This represents the recommended direction for a production-grade ISO 5338 retrieval system.